

Getting Started with Data Analysis and Programming

Python, VS Code, Git, GitHub, Markdown, and Quarto

Summer Semester Undergraduate Data Analysis Course

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1 Class 1 overview

Welcome to data analysis and programming. This course uses **Python** as the main programming language, but the larger goal is not only to “learn Python.” The goal is to learn a way of thinking and working with data:

1. Ask a clear question.
2. Find or collect relevant data.
3. Clean and organize the data.
4. Analyze patterns.
5. Communicate results honestly and clearly.
6. Make the work reproducible so that you, your classmates, or future collaborators can understand what happened.

Students in this class may come from journalism, social science, humanities, business, natural science, engineering, public health, or other fields. Programming is useful across all of these areas because it helps us handle repetitive tasks, large **datasets**, transparent analysis, and reproducible writing.

1.1 Learning objectives

By the end of this first class, students should be able to explain:

- What programming is and why Python is useful for data analysis.
- What an **integrated development environment (IDE)** is.
- What **Visual Studio Code** is and how it helps with coding and writing.
- The difference between **Python**, **VS Code**, and the **Python extension for VS Code**.
- What **Git** and **GitHub** are.
- Why plain-text documents such as **Markdown** and **Quarto .qmd files** are useful for data analysis, writing, and **version control**.
- How Markdown/Quarto differs from traditional word processors such as Microsoft Word.

2 The big picture: tools and concepts

A beginning data analysis workflow often includes several tools:

Tool or concept	What it is	Why we use it in this course
Python	A programming language	To clean, analyze, visualize, and automate work with data
VS Code	A code editor / lightweight IDE	To write Python code, Markdown, Quarto documents, and manage project files
Python extension for VS Code	A VS Code add-on	To make VS Code better for Python: code completion, debugging, environment selection, notebooks, and testing
Git	Version control software	To track changes in files over time
GitHub	A web platform for Git repositories	To store, share, collaborate on, and submit projects
Markdown	A plain-text writing format	To write structured documents that are readable and easy to version-control
Quarto	A publishing system using <code>.qmd</code> files	To combine text, code, results, tables, figures, and references in one reproducible document

Important distinction

VS Code is not Python. VS Code is the place where you write and manage code. Python is the language and **interpreter** that runs the code. The Python extension helps VS Code work well with Python, but it does not replace the need for a Python installation.

3 What is programming?

Programming means writing instructions that a computer can run. A program may be very short or very complex. In data analysis, programs often help us:

- Import a **dataset**.
- Clean messy columns.
- Combine multiple files.
- Count, group, filter, or summarize records.
- Create charts.

- Run statistical models.
- Generate reports.
- Repeat the same analysis when new data arrives.

A simple Python example:

```
# A small Python example
name = "data analysis"
print(f"Welcome to {name}!")
```

```
Welcome to data analysis!
```

Programming is not only about writing code from memory. It is also about reading documentation, asking good questions, testing small pieces, debugging **errors**, and explaining your reasoning.

4 What are programming languages?

A **programming language** is a formal language used to give instructions to a computer. Different languages are often used for different purposes.

Language	Common uses	Notes for beginners
Python	Data analysis, automation, machine learning, web development, scripting	The main language for this course
R	Statistics, data visualization, research reporting	Very common in statistics and social science
JavaScript	Websites, interactive graphics, web apps	Runs in web browsers and servers
SQL	Databases	Used to query structured data
Java	Large applications, Android, enterprise software	Common in computer science courses
C/C++	Systems, performance-critical software, embedded computing	Powerful but less beginner-friendly
Julia	Scientific computing, numerical analysis	Designed for high-performance technical computing

4.1 Why Python for this class?

Python is a good language for an undergraduate data analysis course because it is widely used, readable, and supported by a large ecosystem of data tools. In this course, Python can help us work with:

- Tables and spreadsheets.
- Text data.
- Web data.
- Survey data.
- Public **datasets**.
- **Visualizations**.
- Reproducible reports.

Common Python packages for data analysis include:

Package	Purpose
<code>pandas</code>	Work with table-like data
<code>numpy</code>	Numerical computing
<code>matplotlib</code>	Data visualization
<code>seaborn</code>	Statistical graphics
<code>requests</code>	Download data from the web
<code>beautifulsoup4</code>	Parse HTML from web pages
<code>jupyter</code>	Interactive notebooks and kernels used by many tools, including Quarto Python documents

A small data example:

```
import pandas as pd

students = pd.DataFrame({
    "field": ["Journalism", "Business", "Public Health", "History"],
    "students": [8, 12, 6, 5]
})

students
```

	field	students
0	Journalism	8
1	Business	12
2	Public Health	6
3	History	5

5 What is an IDE?

An **integrated development environment** or **IDE** is software that helps people write, run, test, and manage code.

An **IDE** or code editor may include:

- A file browser.
- Syntax highlighting.
- Code completion.
- Error messages.
- A terminal.
- Debugging tools.
- **Git** integration.
- Extensions for different languages and file formats.

Examples include VS Code, PyCharm, RStudio/Positron, JupyterLab, and Spyder.

Analogy

A programming language is like a language you write in. An **IDE** is like the desk, notebook, dictionary, spell checker, filing cabinet, and workspace that help you write more effectively.

6 What is Visual Studio Code?

Visual Studio Code, often called **VS Code**, is a free code editor made by Microsoft. It is popular because it is flexible, works on Windows/macOS/Linux, supports many languages, and has many extensions.

In this course, VS Code can be used to:

- Write Python scripts: `.py` files.
- Write Quarto documents: `.qmd` files.
- Write Markdown documents: `.md` files.
- Open a project folder.
- Use the built-in terminal.
- Work with **Git** and GitHub.
- Preview Quarto documents.

6.1 Recommended VS Code extensions

Install these extensions from the VS Code Extensions panel:

Extension	Why install it?	Link
Python	Python language support, code completion, debugging,	https://marketplace.visualstudio.com/items?itemName=ms-python.python

Extension	Why install it?	Link
	linting, testing, and environment tools	itemName=ms-python.python
Quarto	Language support and preview/rendering tools for <code>.qmd</code> documents	https://marketplace.visualstudio.com/items?itemName=quarto.quarto
Jupyter	Notebook support inside VS Code	https://marketplace.visualstudio.com/items?itemName=ms-toolsai.jupyter

⚠ Do I still need to install Python if I use VS Code?

Yes. In most teaching setups, students should install **both**:

1. **Python**, the **interpreter** that actually runs Python code.
2. **VS Code + Python extension**, the editor environment that helps write and manage Python code.

The VS Code Python extension is not a complete replacement for Python itself. It helps VS Code find and use an installed Python **interpreter**. After installing Python, VS Code can select that **interpreter** for the project.

7 What is the terminal?

The **terminal** is a text-based way to communicate with your computer. You type commands, and the computer responds.

In data analysis courses, students may use the terminal to:

- Check whether software is installed.
- Run Python scripts.
- Install Python packages.
- Use **Git** commands.
- Render Quarto documents.

Common commands:

```
python --version
pip --version
git --version
quarto --version
```

To install Python packages, students may use:

```
pip install pandas matplotlib jupyter
```

For a reproducible course setup, you may later introduce virtual environments, such as:

```
python -m venv .venv
```

8 What is Git?

Git is a **version control** system. It records changes to files over time.

Git helps answer questions such as:

- What changed since yesterday?
- Who changed this file?
- Can I go back to an earlier version?
- Can two people work on the same project without emailing files back and forth?
- Can I show the history of my analysis?

8.1 Key Git vocabulary

Term	Meaning
Repository / repo	A project folder tracked by Git
Commit	A saved snapshot of changes
Commit message	A short explanation of what changed
Branch	A separate line of work
Merge	Combining changes from different branches
Clone	Copying a repository from a remote location to your computer
Pull	Bringing remote changes to your computer
Push	Sending local commits to a remote repository
<code>.gitignore</code>	A file that tells Git which files not to track

8.2 A simple Git workflow

```
git status
git add analysis.qmd
git commit -m "Add first analysis draft"
git push
```

In VS Code, many **Git** actions can also be done through the Source Control panel without typing every command manually.

9 What is GitHub?

GitHub is a web platform for hosting **Git** repositories. **Git** is the **version control** tool; **GitHub** is a website that makes **Git**-based collaboration easier.

GitHub can be used to:

- Store a project online.
- Share code and documents.
- Submit assignments.
- Collaborate with classmates.
- Review changes.
- Track issues and tasks.
- Publish project websites.

i Git vs GitHub

Git and **GitHub** are related, but they are not the same thing.

- **Git** is software installed on your computer.
- **GitHub** is an online service that hosts **Git** repositories.

10 Why writing matters in a data analysis course

Data analysis is not complete when the code runs. Students also need to explain:

- What question they asked.
- Where the data came from.
- What assumptions they made.
- What methods they used.
- What results they found.
- What limitations remain.
- What readers should conclude.

For students from journalism and other writing-intensive fields, this is especially important. A good data project combines evidence, computation, and clear writing.

11 What is Markdown?

Markdown is a lightweight plain-text format for writing structured documents. A **Markdown** file can be opened in any text editor, and the file remains readable even before it is converted to **HTML**, **PDF**, or **Word**.

Common **Markdown** syntax:

```
# Heading 1
```



```

## Heading 2

This is a paragraph with bold text and italic text.

- Item one
- Item two
- Item three

[Link text](https://example.com)

| Column A | Column B |
|---|---|
| Value 1 | Value 2 |

```

Markdown is especially useful because it stores writing as plain text. Plain text works well with **Git**, because Git can clearly show exactly which lines changed.

12 What is Quarto?

Quarto is an open-source publishing system for technical and scientific documents. A Quarto file usually has the extension `.qmd`.

A `.qmd` file can include:

- Markdown text.
- Python code blocks.
- Tables.
- Figures.
- Citations.
- Equations.
- Callout boxes.
- Links.
- Document metadata.

Quarto can render one source file into multiple output formats, such as HTML, PDF, Word, slides, websites, books, and dashboards.

12.1 A minimal Quarto document

```

---
title: "My First Quarto Document"
format: html
jupyter: python3
---

```

```
# Introduction

This document combines writing and code.

::: {.cell execution_count=3}
``` {.python .cell-code}
print("Hello, Quarto!")
```

::: {.cell-output .cell-output-stdout}
```
Hello, Quarto!
```

:::
:::
```

13 Why Markdown and Quarto instead of only Microsoft Word?

Microsoft Word is useful and familiar. It is often the right tool for polished documents, tracked comments, administrative forms, and situations where collaborators expect `.docx` files.

However, for data analysis and programming, Markdown and Quarto have several advantages.

| Need | Microsoft Word | Markdown / Quarto |
|------------------------------|---|---|
| Human-readable source file | <code>.docx</code> is a complex document format | Plain text; readable in any editor |
| Version control with Git | Difficult to see exact line-by-line changes | Excellent; Git can show precise text/code changes |
| Combining code and writing | Usually requires copying results manually | Code, output, and explanation can live in one document |
| Reproducibility | Charts/tables may be pasted manually | Results can be regenerated from code |
| Collaboration through GitHub | Possible, but not natural | Designed for text-based collaboration |
| Multiple output formats | Mainly Word/PDF depending on workflow | Same source can render to HTML, PDF, Word, slides, websites, etc. |

| Need | Microsoft Word | Markdown / Quarto |
|-----------------------|---------------------------------|---|
| Learning curve | Familiar interface | Requires learning simple syntax and tools |
| Visual layout control | Strong visual page layout tools | Better for structured writing than fine-grained visual design |

13.1 Example: why version control is easier with plain text

Imagine a student changes this sentence:

```
The city budget increased by 4% in 2024.
```

Later, the sentence becomes:

```
The city budget increased by 6% in 2024 after revised revenue estimates.
```

Git can show the exact line that changed. In a Word document, version history exists, but the internal file format is not as simple for Git to compare, merge, and review.

13.2 When should students use Word?

Word is still useful when:

- The instructor, editor, or organization requires `.docx`.
- A document needs detailed visual layout.
- A collaborator is not comfortable with Markdown, Quarto, or **Git**.
- The task is mainly final formatting rather than reproducible analysis.

13.3 When should students use Quarto?

Quarto is especially useful when:

- A document includes code and results.
- The analysis may need to be repeated.
- The project benefits from **Git** version history.
- The writer wants to produce HTML, PDF, Word, or slides from one source file.
- The document should clearly connect evidence, method, and conclusion.

Teaching message

For this course, students do not need to reject Word. Instead, they should learn that different tools serve different purposes. Word is strong for conventional document editing. Markdown and Quarto are strong for transparent, reproducible, version-controlled analytical writing.

14 Suggested installation checklist

Students should install or create accounts for the following tools before the second class.

| Step | Tool | What to do | Link |
|------|--------------------------|---|---|
| 1 | Python | Download and install Python 3 | https://www.python.org/downloads/ |
| 2 | VS Code | Download and install Visual Studio Code | https://code.visualstudio.com/Download |
| 3 | VS Code Python extension | Install inside VS Code | https://marketplace.visualstudio.com/items?itemName=ms-python.python |
| 4 | Git | Download and install Git | https://git-scm.com/downloads |
| 5 | GitHub | Create a GitHub account | https://github.com/ |
| 6 | Quarto | Download and install Quarto | https://quarto.org/docs/download/ |
| 7 | Quarto VS Code extension | Install inside VS Code | https://marketplace.visualstudio.com/items?itemName=quarto.quarto |
| 8 | Jupyter package | Install with Python package manager | Run <code>pip install jupyter</code> in the terminal |

Optional but useful:

| Tool | Purpose | Link |
|-------------------------|--|---|
| GitHub Desktop | A graphical way to use GitHub and Git | https://desktop.github.com/ |
| GitHub Docs: Git basics | Beginner-friendly Git/GitHub documentation | https://docs.github.com/en/get-started/git-basics |
| Quarto Markdown basics | Markdown syntax used in Quarto | https://quarto.org/docs/authoring/markdown-basics.html |

| Tool | Purpose | Link |
|------------------------------|--------------------------------------|---|
| Quarto with VS Code tutorial | Official Quarto tutorial for VS Code | https://quarto.org/docs/get-started/hello/vscode.html |

15 Recommended first-week setup test

After installing Python, VS Code, **Git**, and Quarto, students can open the terminal in VS Code and run:

```
python --version
git --version
quarto --version
```

Then install basic Python packages:

```
pip install pandas matplotlib jupyter
```

Then create a file called `hello.qmd`:

```
---
title: "Hello Data Analysis"
format: html
jupyter: python3
---

# My first Quarto document

This is a paragraph written in Markdown.

::: {.cell execution_count=4}
``` {.python .cell-code}
print("Hello from Python inside Quarto!")
```

::: {.cell-output .cell-output-stdout}
```
Hello from Python inside Quarto!
```

:::
:::
```

Render the document in VS Code using the Quarto extension, or from the terminal:

```
quarto render hello.qmd
```

16 Suggested in-class activity

16.1 Activity: compare three ways of working

Ask students to work in small groups and compare these workflows:

1. Writing analysis notes in Microsoft Word.
2. Writing notes in Markdown.
3. Writing a Quarto document with text and Python code together.

Discussion questions:

- Which workflow feels most familiar?
- Which workflow makes the analysis easiest to reproduce?
- Which workflow would work best for a journalism story with charts?
- Which workflow would work best for a research assignment?
- Which workflow makes it easiest to see what changed between drafts?

17 Common beginner questions

17.1 Is VS Code the same as Visual Studio?

No. **Visual Studio Code** is a lightweight editor used for many languages and workflows. **Visual Studio** is a larger development environment often used for software engineering in languages such as C# and C++. This course uses VS Code.

17.2 Is GitHub the same as Google Drive or Dropbox?

No. GitHub stores project history using **Git**. It is not just file storage. It tracks changes, supports collaboration, and lets people review project history.

17.3 Can I use Word and Quarto together?

Yes. Quarto can render documents to Word format. A common workflow is to write a reproducible analysis in Quarto, then export to Word when a professor, editor, or organization requires `.docx`.

17.4 Do I need to memorize terminal commands?

No. Students should learn a small set of common commands, but the main goal is understanding the workflow. VS Code also provides graphical tools for many tasks.

17.5 Why not use only Jupyter notebooks?

Jupyter notebooks are useful for exploration and interactive coding. Quarto is useful for turning analysis into polished documents, reports, websites, or slides. In this course, we may use both, but `.qmd` files are especially helpful for combining writing, code, and **version control**.

18 One-sentence summaries

- **Python** is the language we use to analyze data.
- **VS Code** is the editor where we write code and documents.
- **The Python extension** helps VS Code work well with Python, but it does not replace Python itself.
- **Git** tracks changes to project files.
- **GitHub** hosts **Git** projects online and supports collaboration.
- **Markdown** is a plain-text writing format.
- **Quarto** turns Markdown plus code into reproducible documents.
- **Word** is useful for traditional document editing, while **Quarto** is useful for reproducible, version-controlled analytical writing.

19 Instructor notes

Possible teaching sequence for a 75-minute first class:

| Time | Topic |
|-----------|--|
| 0–10 min | Why data analysis and programming matter across fields |
| 10–20 min | Programming languages and why this course uses Python |
| 20–30 min | IDEs, VS Code, extensions, and the terminal |
| 30–45 min | Git, GitHub, and version control |
| 45–60 min | Markdown, Quarto, and writing reproducible documents |
| 60–70 min | Setup checklist and demonstration |
| 70–75 min | Questions and next steps |

Possible first assignment:

1. Install Python, VS Code, **Git**, and Quarto.
2. Create a GitHub account.
3. Make a folder for the course.
4. Create a `hello.qmd` file.
5. Render it to HTML.
6. Submit the `.qmd` file and rendered HTML file, or submit a GitHub repository link if GitHub is already introduced.